RadeonShift AI

Accelerating the MI300X Hardware Migration via Generative AI

The Problem: CUDA Lock-in

Enterprise AI workloads are facing massive bottlenecks due to a reliance on legacy NVIDIA CUDA codebases.

- Manually migrating a 50,000-line CUDA codebase to ROCm takes **6+ months** of specialized engineering time.
- Portability tools exist (e.g. `hipify-perl`), but they lack the architectural awareness to optimize code for AMD Instinct architecture (like 64-wide wavefronts).
- Teams lack visibility into whether their migrated code will actually compile and run on AMD hardware without direct bare-metal access.

The Solution: RadeonShift AI

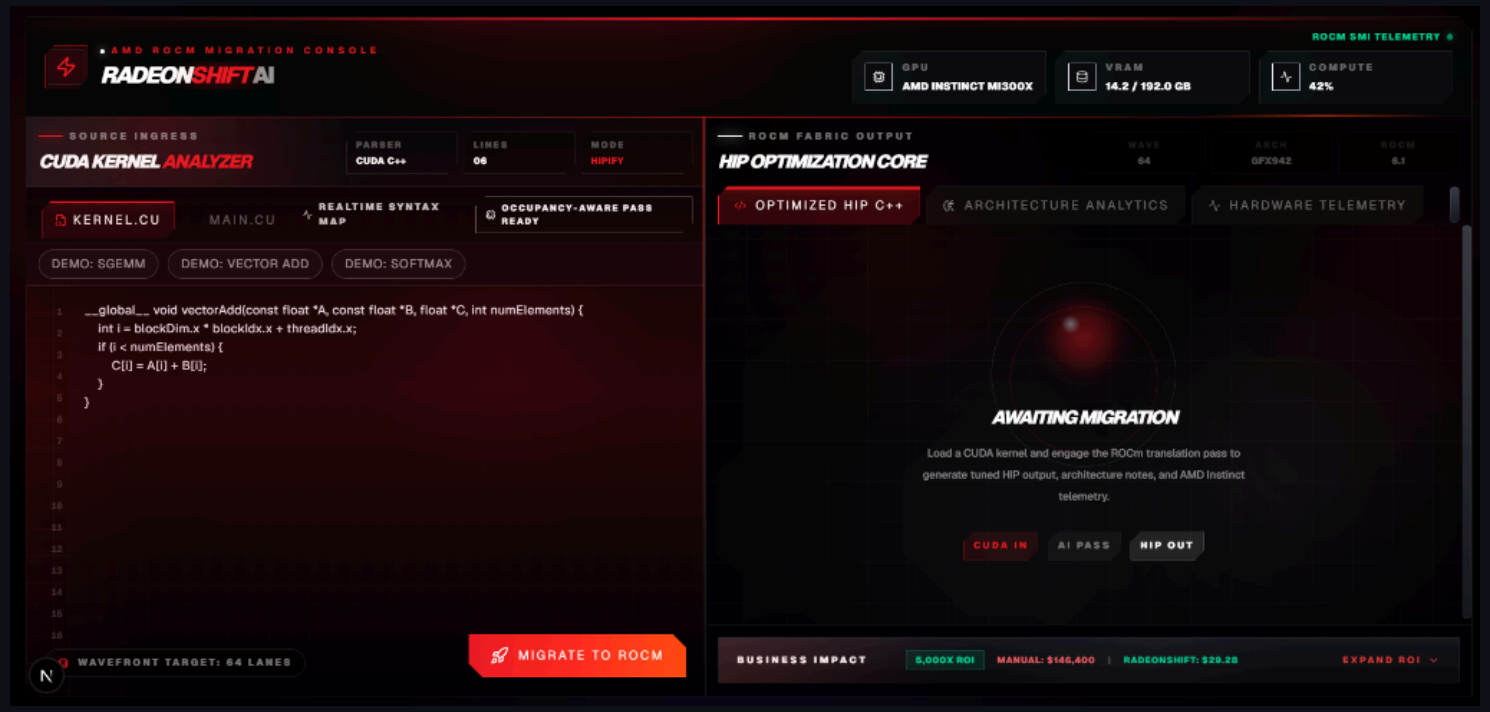
RadeonShift AI is an AI-assisted CUDA kernel migration assistant that translates CUDA to HIP/ROCm and audits architecture-specific migration risks.

1. **Syntax Translation:** Uses Fireworks-hosted translation model for AI Translation.
2. **MoA Audit:** Leverages DeepSeek-V4 to scan for PTX risks, hardcoded warp sizes, and AMD-specific optimizations.
3. **Optional Hardware Evidence:** When the ROCm backend is online, RadeonShift can compile-check generated HIP and show MI300X telemetry. Benchmark mode uses trusted internal kernels and clearly labels live, cached, or unavailable evidence.

Architecture

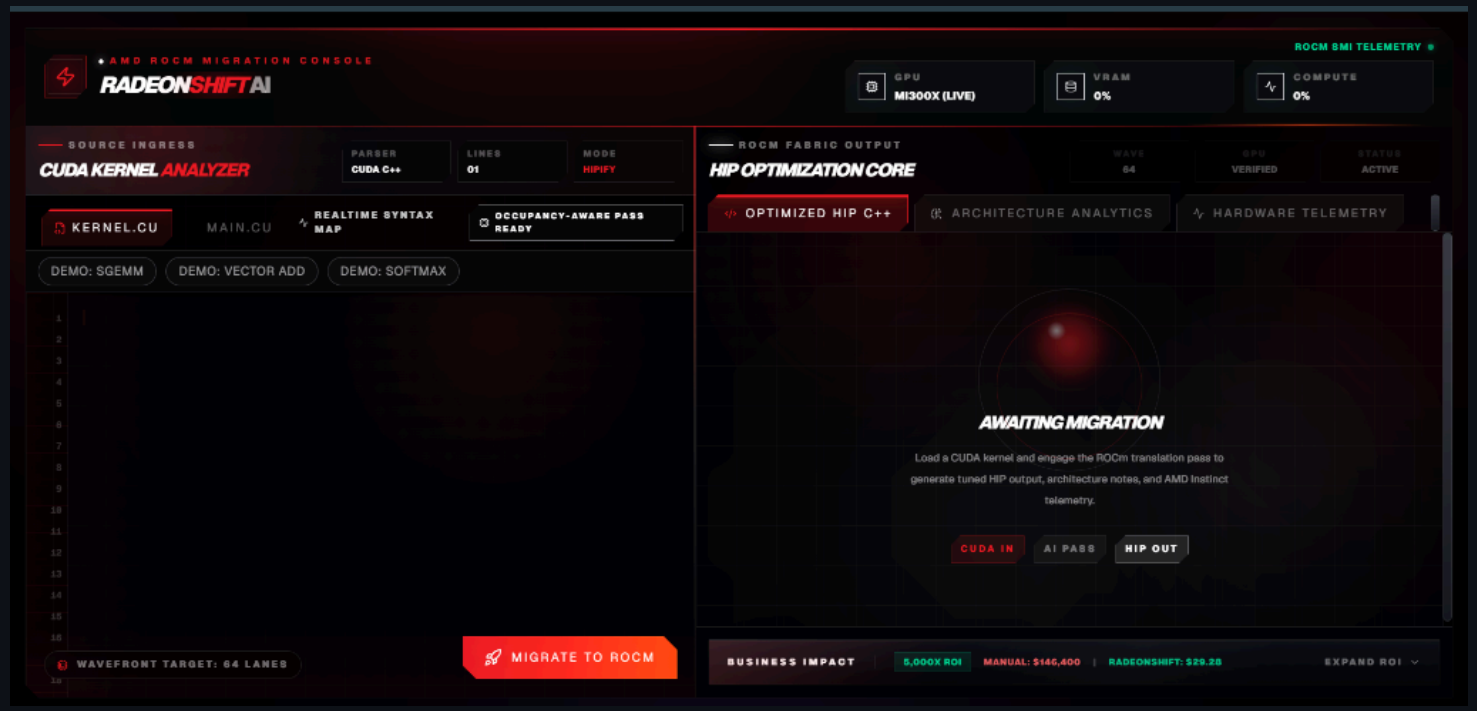
1. **Frontend (Next.js):** Manages user state and renders the dynamic dashboard.
2. **AI Translation Layer (Vercel / Next.js API Routes):** Orchestrates AI translation and Mixture-of-Agents audit via Fireworks AI, operating independently of hardware availability.
3. **Optional Hardware Engine (FastAPI via Pinggy):** A bare-metal ROCm layer that can compile-check generated HIP via `hipcc`, poll `rocm-smi` for telemetry, and run trusted benchmark kernels when connected.

Demo: The RadeonShift Dashboard



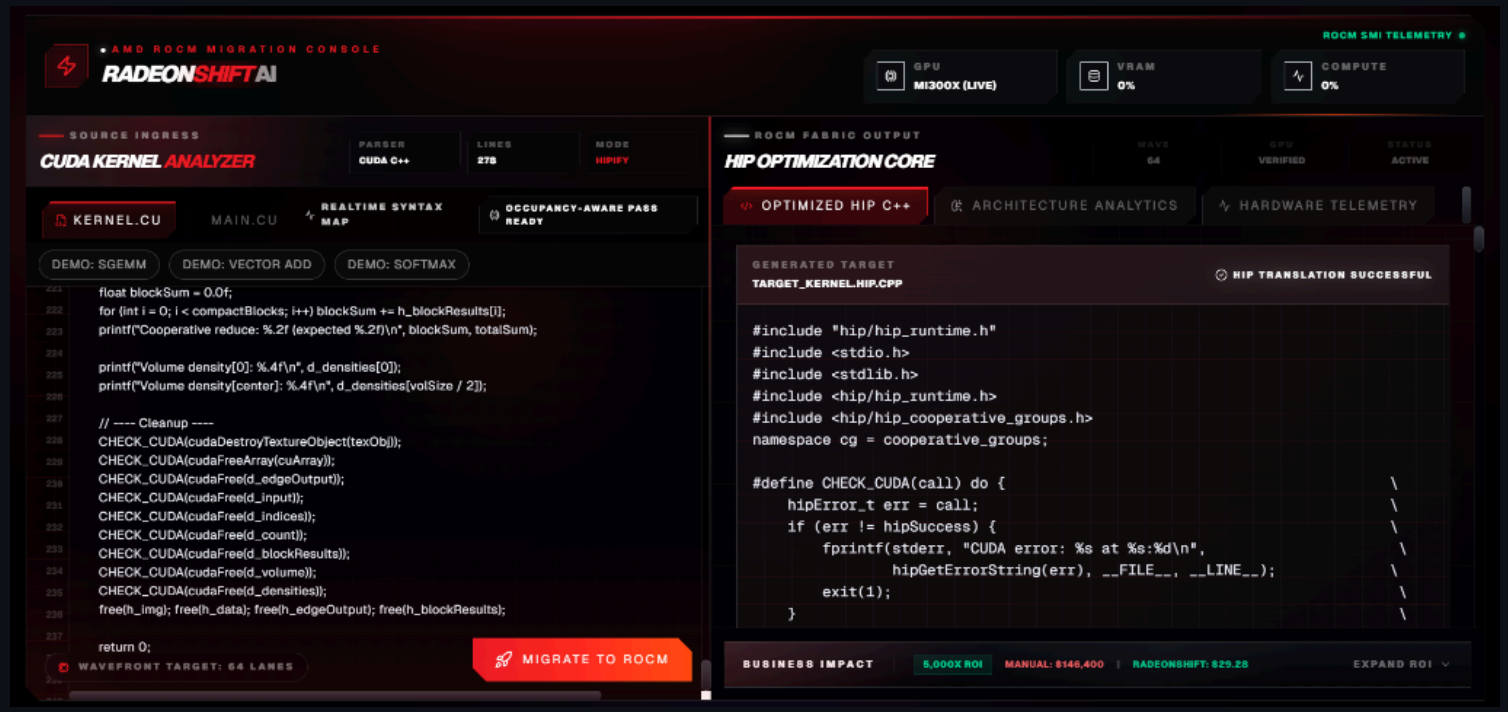
- **Source Code Editor:** Paste raw CUDA C++ kernels.
- **HIP Optimization Core:** View the translated target kernel.
- **MoA Audit Scorecard & Universal Review:** See readiness scores, live PTX/wavefront findings, and auto-patch suggestions.

Step 1: The Awaiting Migration State



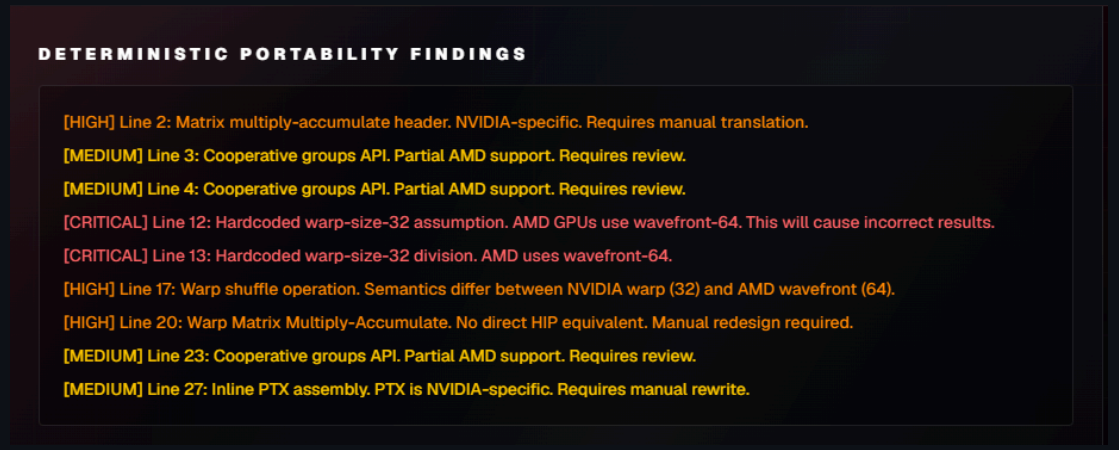
Before translation begins, the platform awaits the CUDA payload. The user inputs their native NVIDIA code into the **CUDA Kernel Analyzer**. The system prepares for the AI-assisted hardware pass.

Step 2: HIP Optimization Core



Upon engaging the ROCm translation pass, the core converts the syntax. The resulting C++ HIP code (`target_kernel.hip.cpp`) is immediately presented, demonstrating generated HIP translation.

Step 3: Architecture Analytics — Screenshots



Step 3: Architecture Analytics — Two Layers

The MoA Audit Scorecard evaluates code through two distinct layers:

Layer 1 — Deterministic Risk Detection:
Hardcoded rule-based scanners check for known AMD portability risks before any AI analysis runs.

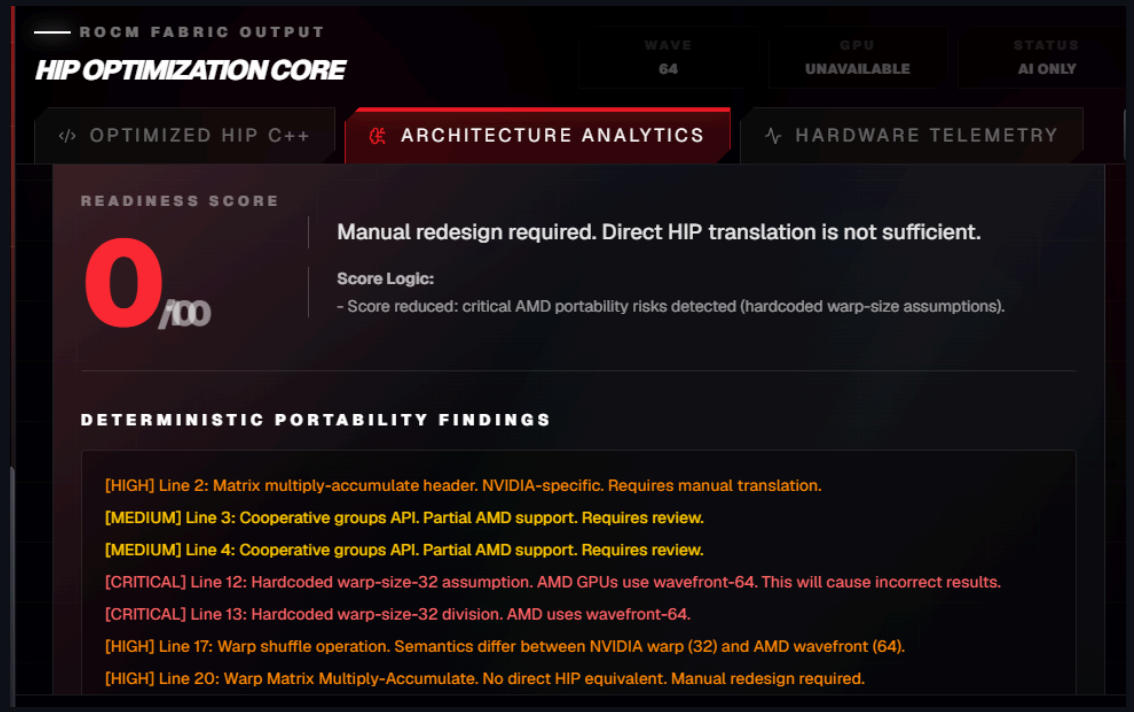
Layer 2 — AI-Powered Analysis:
The MoA pipeline then runs semantic analysis for deeper architectural risks the deterministic layer cannot catch alone.

Step 3: Deterministic Patterns Detected

- Hardcoded warp-size assumptions (`% 32` , `/ 32`) → **CRITICAL**
- Warp shuffle operations (`__shfl_sync` , `__shfl_up` , `__shfl_down` , `__shfl_xor`) → **HIGH**
- WMMA / matrix multiply-accumulate (`wmma` , `mma.h`) → **HIGH**
- Async memory copy (`cuda::memcpy_async`) → **HIGH**
- Cooperative groups → **MEDIUM**
- Inline PTX assembly (`asm`) → **MEDIUM**

Both layers are displayed separately in the report — full transparency into rules-based vs. AI findings.

Step 3a: Deterministic Redesign Guardrails



RadeonShift's Deterministic Rules Engine catches unsupported architectures before AI translation is trusted. Findings appear in a dedicated "**Deterministic Portability Findings**" section, independent of AI output.

Step 3a: Correctness over Completeness

Safely enforces **MANUAL REDESIGN REQUIRED** if code relies on hardware-specific features that have no direct HIP equivalent.

Score Capping:
Drops readiness score to **< 50%** automatically when CRITICAL or HIGH deterministic findings are present — discouraging unsafe architecture conversions from being treated as ready to ship.

Step 3a: Explainable Score Logic

The score now includes a **visible explanation line** showing exactly why the confidence was reduced:

- "*Score reduced: critical AMD portability risks detected*"
- "*Score capped: unsupported CUDA features require manual redesign*"
- "*Flagged for review: NVIDIA-specific constructs detected*"
- "*No deterministic portability risks detected*" (when clean)

Judges can trace **every score change** to a specific deterministic or AI finding.

Step 4: Live Hardware Telemetry

When the optional backend is online, the platform connects to a remote bare-metal **AMD MI300X** environment and surfaces live ROCm telemetry plus compile-check evidence.

When offline, the platform enters **Demo Mode** — a guaranteed static sample result is loaded containing a complete wavefront-bug detection flow, so the full product can always be demonstrated regardless of backend status.

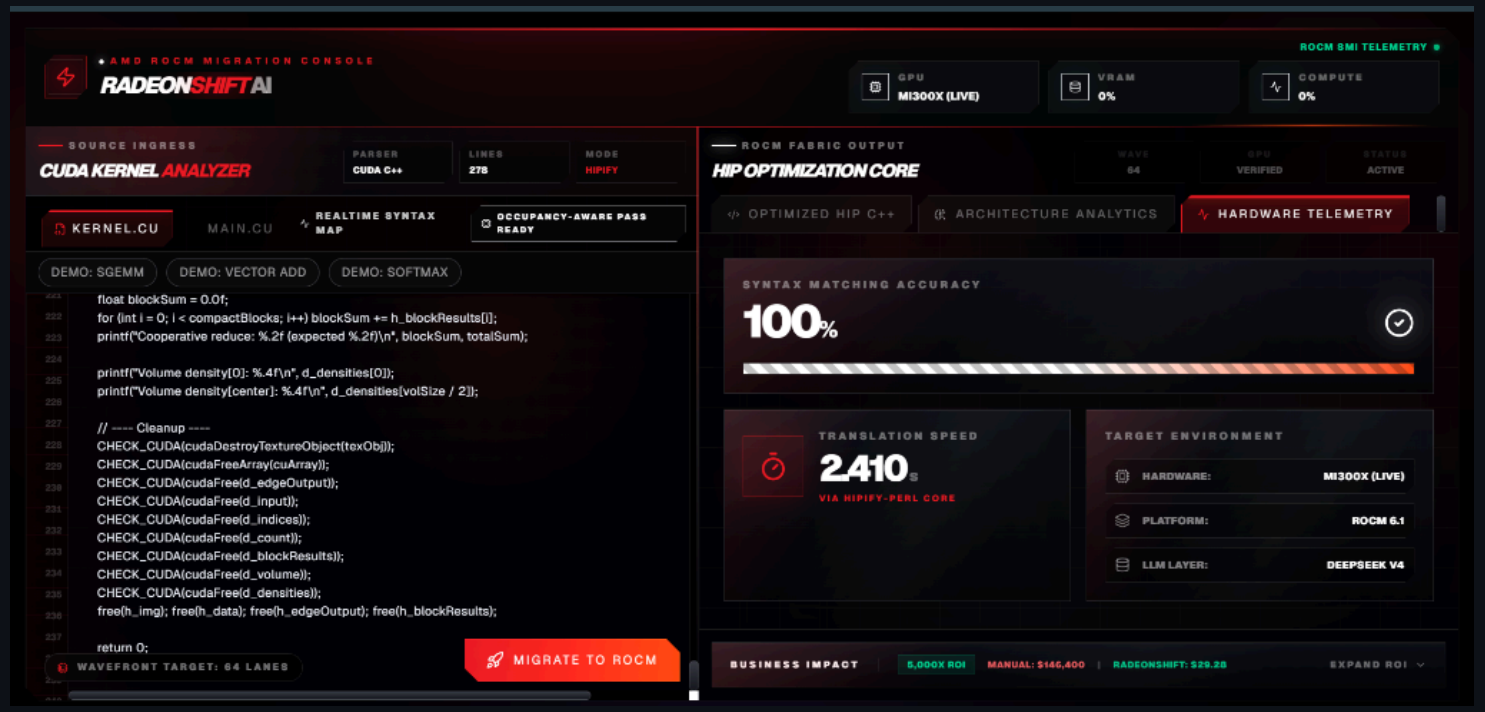
Step 4: Transparency & Provenance

Three transparency states are clearly labeled in the UI:

- 🟢 **LIVE:** Real-time MI300X telemetry active
- 🟡 **CACHED:** Previously fetched evidence displayed with timestamp
- 🟠 **DEMO MODE:** Static guaranteed sample loaded, clearly labeled as demo data

No state is ever hidden or ambiguous — the provenance of every metric is always **visible**.

Step 5: Bare-Metal Benchmark Execution



Benchmark mode runs trusted HIP benchmark kernels on MI300X when hardware is online. RadeonShift separates AI audit, compile-check evidence, and runtime benchmark provenance — nothing is fabricated.

Business Value & ROI

Illustrative ROI Model

- **Manual Migration:** 244 Kernels × 4 hrs = 976 Engineer-Hours (~$146,000, based on $150/hr senior GPU engineer rate)
- **RadeonShift Migration:** 244 Kernels × illustrative ~$0.12 AI/compute cost = ~$29 first-pass triage estimate
- **Time Savings:** Initial audit and translation can shrink from weeks of first-pass review to minutes, with human validation still required for production.

Trust & Market Size

New — Trust Through Deterministic Guardrails:
Unlike AI-only translation tools, RadeonShift's deterministic rules engine catches critical AMD portability risks before AI output is trusted — reducing false-confidence migrations and ensuring engineers know exactly which kernels need manual redesign.

TAM: $4.2B illustrative GPU migration and porting services market estimate.

The AMD Infrastructure Story

- **Hardware Target:** Optimized for AMD Instinct MI300X
- **Software Stack:** Built on ROCm 6.x APIs and `hipcc`
- **AI Acceleration:** MoA pipeline runs through the Fireworks-hosted inference API
- **Deterministic Engine:** Standalone rule-based portability scanner runs independently of AI, catching hardcoded warp assumptions, WMMA, async-copy, and PTX risks
- **Honest Fallbacks:** If the backend lacks ROCm hardware, the platform enters Demo Mode with a guaranteed static sample, explicitly labeling all evidence as demo data without fabricating live metrics

Engineering Retrospective (1/2)

Bridging a Serverless Frontend with Bare-Metal Hardware

1. **Live Telemetry:** Defeated Next.js caching via `force-dynamic` to stream real-time MI300X metrics.
2. **Defensive APIs:** Built robust frontend parsing to gracefully handle sudden JSON schema changes.
3. **CORS Routing:** Bypassed strict mixed-content blocks by tunneling through Pinggy and Next.js `rewrites()`.
4. **Transparent Debugging:** Overhauled error handling to surface remote Python stack traces in the UI.

Engineering Retrospective (2/2)

Adding Deterministic Safety & Demo Stability

5. **Structured AI Prompts:** Constrained LLM outputs to raw HIP code or JSON audit findings, reducing UI parsing failures.
6. **Deterministic + AI Separation:** Built a standalone rules engine that runs before AI analysis — judges can distinguish rule-based detection from AI inference.
7. **Explainable Score Logic:** Added visible score explanation lines so every confidence change is traceable.
8. **Demo-Safe Fallback:** Built a guaranteed static sample mode so the full product flow can always be demonstrated even without bare-metal hardware.

Closing & Team

RadeonShift AI is bridging the gap between legacy NVIDIA codebases and the future of AMD compute — with deterministic guardrails that ensure every migration is safe, explainable, and honest.

- **GitHub Repository:** shashankh3/RadeonShift-AI
- **Live Demo:** radeon-shift-ai.vercel.app

Thank You!

Shashank Hirwani
Unknown Hacker (shashankh366207)